Introduction to Large Language Models
Assignment- 6

TYPE OF QUESTION: MCQ
Number of questions: 8 **Total mark: 6 X 1 + 2 X 2 = 10**

QUESTION 1: [1 mark]

State True or False.

Rotary Position Embedding (RoPE) encodes position by applying a rotation to the query and key vectors rather than by adding a positional vector to the input embedding.

a. True
b. False

Correct Answer: a

Solution: RoPE is multiplicative: it rotates query and key vectors by an angle proportional to their position, so the dot product between them depends on their relative offset. Sinusoidal positional encoding, by contrast, is additive.

QUESTION 2: [1 mark]

Which of the following is true about multi-head attention?

a. All heads share a single set of attention weights, which improves interpretability.
b. Each head uses its own learned projections and can attend to different representation subspaces in parallel.
c. It reduces the total number of parameters relative to single-head attention of the same width.
d. The heads are averaged before the softmax is applied.

Correct Answer: b

Solution: Each head has independent Q, K and V projections, so different heads can specialise on different kinds of relationships. Their outputs are concatenated and passed through an output projection.

QUESTION 3: [1 mark]

What is the role of the residual connection in a Transformer block?

a. It improves gradient flow during backpropagation and makes deep stacks trainable.
b. It normalises the input embeddings.
c. It reduces the computational complexity of attention.
d. It removes the need for positional information.

Correct Answer: a

Solution: Please refer to lecture slides.

QUESTION 4: [1 mark]

What is the primary reason Transformers use layer normalization instead of batch normalization?

a. Layer normalization is always faster to compute than batch normalization.
b. Batch normalization is not effective for sequence models, where statistics vary with batch composition and sequence length.
c. Layer normalization by itself prevents overfitting.
d. Batch normalization is incompatible with positional encodings.

Correct Answer: b

Solution: Layer normalization computes statistics per token across the feature dimension, so it is independent of batch size and of padding, unlike batch normalization which depends on the composition of the batch.

QUESTION 5: [1 mark]

Which of the following statements is NOT true?

a. Self-attention allows every token to attend to every other token in the sequence.

b. Multi-head attention increases the output dimensionality of the Transformer block relative to a single head of the same total width.

c. Relative positional encoding captures the distance between tokens rather than their absolute positions.

d. Positional encoding is necessary because self-attention on its own is permutation invariant.

Correct Answer: b

Solution: The per-head dimension is d_model / h, so concatenating all heads returns a vector of size d_model. Multi-head attention changes how the representation is computed, not its dimensionality.

QUESTION 6: [1 mark]

How does masked self-attention support autoregressive generation?

a. By allowing every position to attend to all tokens, including future ones.

b. By restricting each position to attend only to itself and earlier positions, preventing information leakage from the future.

c. By discarding positional information within the sequence.

d. By disabling the attention mechanism during training.

Correct Answer: b

Solution: Causal masking sets the attention scores of future positions to negative infinity before the softmax, so their weights become 0 and the prediction at each position depends only on the tokens before it.

QUESTION 7: [2 marks]

A self-attention layer has input dimension d_model = 1024 and uses 16 heads. What is the dimension of the output of each individual head?

a. 64

b. 128

c. 32

d. 256

Correct Answer: a

Solution: Each head operates on d_model / h = 1024 / 16 = 64 dimensions. The 16 head outputs are concatenated back into a 1024-dimensional vector.

QUESTION 8: [2 marks]

For a Transformer with d_model = 512, compute the positional encoding for position p = 20 at dimensions 4 and 5, using the sinusoidal formulation:

PE(p, 2i) = sin(p / 10000^(2i / d_model)) PE(p, 2i+1) = cos(p / 10000^(2i / d_model))

a. sin(20 / 10000^(1/128)), cos(20 / 10000^(1/128))
b. cos(20 / 10000^(1/128)), sin(20 / 10000^(1/128))
c. sin(20 / 10000^(4/512)), cos(20 / 10000^(5/512))
d. sin(20 / 10000^(1/256)), cos(20 / 10000^(1/128))

Correct Answer: a

Solution: Dimensions 4 and 5 form the pair with 2i = 4, so i = 2 and both use the same frequency term 2i / d_model = 4 / 512 = 1 / 128.
PE(20, 4) = sin(20 / 10000^(4/512)) = sin(20 / 10000^(1/128))
PE(20, 5) = cos(20 / 10000^(4/512)) = cos(20 / 10000^(1/128))
